

ROBIN BILODEAU

Ottawa, Ontario, Canada

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EDUCATION

University of Ottawa

Ottawa, Ontario, Canada

Bachelor of Science in Computer Science and Mathematics

Sep 2024 – Apr 2028

- **Relevant Courses:** Intro to Computing 1, Intro to Computing 2, Linear Algebra, Differential Equations and Laplace Transforms, Discrete Structures, Elementary Real Analysis, Introduction to probability

WORK EXPERIENCE

TutorBright

Ottawa, Ontario, Canada

Tutor Mentor

Feb 2026 – Present

- Tutoring students in mathematics, and computer science, helping them improve their understanding and performance in these subjects

Scotiabank - Global Banking and Markets

Toronto, Ontario, Canada

Quantitative Analyst Intern

Sep 2025 – Dec 2025

- Refactored a legacy pricing process into a modular architecture supporting multiple rates derivatives, including swaps, swaptions, caps, and floors
- Optimized performance bottlenecks in core analytical functions, leading to substantial runtime improvements
- Conducted in-depth testing and investigation of pricing models to identify product-dependent discrepancies and assess model validity, and visualized discrepancy differences across products to communicate findings clearly

RESEARCH EXPERIENCE

Uncertainty-Aware Statistical Extensions for HDBSCAN Clustering

Ottawa, Ontario, Canada

Undergraduate Researcher

Apr 2025 – Oct 2025

- Developed a Monte Carlo sampling framework to repeatedly adjust probability rank matrices, yielding statistically stable core-distance estimates for clustering high-noise data leading to reduced sensitivity to outliers and improved cluster detection reliability
- Coordinated a fast Fourier-based deconvolution estimator to recover underlying data distributions from noise-contaminated observations, enhancing core distance estimates for robust HDBSCAN clustering

PROJECTS

OrderBook Implementation for Financial Markets

- Built a high-performance real-time order book system with FastAPI, supporting MARKET, LIMIT, and FOK orders, handling order placement, matching, and cancellations
- Leveraged Python's bisect module to maintain sorted price levels in the order book, significantly improving insert/search performance over naive list operations
- Designed robust order validation using StrEnum to enforce consistent order types (LIMIT, MARKET, FOK) and sides (BUY, SELL), enhancing code clarity and reducing bugs

LEADERSHIP AND ACTIVITIES

SSC Case Study Competition - Parkinson's Disease Classification

Hamilton, Ontario, Canada

Team Member

Apr 2026 – Jun 2026

- Performed end-to-end data cleaning on a high-dimensional clinical dataset (500+ variables) from the Canadian Open Parkinson Network (C-OPN), including leakage analysis to identify and remove variables that directly encode diagnostic criteria for PD vs. atypical parkinsonism, and outcome-stratified MICE imputation to handle missing data
- Developed a random forest classifier achieving greater than 95% accuracy in distinguishing Parkinson's disease from atypical parkinsonism syndromes, using stratified k-fold cross-validation with fold splitting prior to imputation

SKILLS, LANGUAGES, INTERESTS

- **Languages:** English (Native speaker), French (Native speaker)
- **Programming:** Python, R, Java, Typst, C++ (Beginner)
- **Tools:** Microsoft Word, Git, Visual Studio Code, Canva, Jupyter
- **Libraries:** matplotlib, numpy, Scikit-learn, pandas, yfinance
- **Athletics:** Boxing, Hockey, Soccer, Volleyball
- **Clubs:** Mathematics Club, Undergraduate Research Club